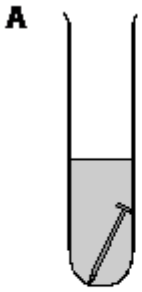
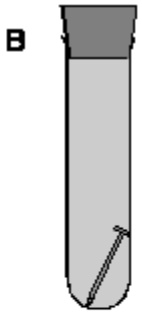
Q1.

Jessica was investigating the rusting of iron. She set up five experiments as shown below, and left the test-tubes for three days.



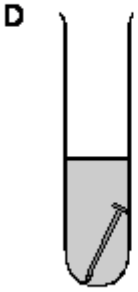
iron nail in distilled water



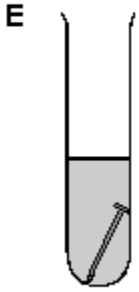
iron nail in tap water which has been boiled to remove dissolved gases



iron nail and a chemical to absorb water vapour



iron nail in sea water



iron nail in vinegar

Jessica wrote the following results in her book.

Test–tube	observation
A	nail slightly rusty
B	nail still shiny
C	nail still shiny
D	nail very rusty
E	nail slightly rusty, bubbles of gas seen

(a) Explain why the nails had **not** rusted in test-tubes B and C.

in test-tube B …….

……..

in test-tube C …….

……..

2 marks

(b) In test-tube E the iron nail reacted with the vinegar.

(i) Is vinegar **acidic**, **alkaline** or **neutral**?

…….

1 mark

(ii) When the iron reacted with the vinegar, bubbles of gas were formed.
What gas was formed?

…….

1 mark

(c) Before putting the iron nail in test-tube D, Jessica weighed the nail.
After three days she dried and weighed the nail **and** the rust which had formed.

(i) How did the total mass of the nail and rust compare to the mass of the nail at the beginning?

……

1 mark

(ii) Give the reason for your answer.

……

……

1 mark

(d) Jessica concluded that the presence of salt in the water made the nail rust more quickly.
Explain why she drew that conclusion from her experiments.

……..

……..

……..

1 mark
Maximum 7 marks

Q1.

(a) in tube B: no oxygen

1 (L5)

in tube C: no water **or** water vapour

accept 'no air'

accept 'no moisture'

accept 'it was dry' or 'it was not wet'

1 (L5)

(b) (i) acidic

1 (L5)

(ii) hydrogen

1 (L6)

(c) (i) it increased **or** it was more

accept 'it was heavier'

1 (L6)

(ii) any **one** from

- oxygen **or** water was added
- the oxygen has mass
- rust contains iron and oxygen **or** water

  accept 'rust is iron oxide'
- the iron reacted with oxygen **or** water

1 (L6)

(d) **Answers must refer to either test-tube D or to sea water.**

any **one** from

- the nail was more rusty in D than in A

  accept 'D was the only one which was rusty'

  accept 'D was very rusty'
- it was more rusty in sea water
- sea water contains salt

1 (L6)

[7]